

Section 4.7 SELECTION OF PROPER CONNECTION HARDWARE



WARNING

Always connect roundslings to properly sized and rated fittings and / or materials

4.7.1 For polyester roundslings, connection hardware should be selected such that it either:

- Conforms to the size requirements listed in Tables 4-6 and 4-7,

OR

- Sized such that the bearing stress value at the connection does not exceed 7,000 Lbs./in² during sling loading. (See calculations below).

Table 4-6 Suitable Connection Hardware Sizes for Polyester Roundslings, When Used in either a Vertical or Choker Hitch

Roundsling		Hardware Size			
WSTDA Roundsling Size	Rated Capacity - Vertical Hitch (Lbs.)	Minimum Stock Diameter or Thickness (Inches)	Minimum Stock Diameter or Thickness (Inches)* ²	Minimum Effective Contact Width* ³ (Inches)	Minimum Effective Contact Width* ³ (Inches) * ²
1	2,600	.39	7/16	.97	1
2	5,300	.59	5/8	1.29	1 3/8
3	8,400	.72	3/4	1.66	1 3/4
4	10,600	.85	7/8	1.78	1 7/8
5	13,200	.95	1	2.00	2
6	16,800	1.12	1 1/8	2.13	2 1/8
7	21,200	1.15	1 3/16	2.62	2 5/8
8	25,000	1.25	1 1/4	2.85	2 7/8
9	31,000	1.41	1 1/2	3.15	3 1/4
10	40,000	1.60	1 5/8	3.57	3 5/8
11	53,000	1.90	2	4.00	4
12	66,000	2.05	2 1/8	4.60	4 5/8
13	90,000	2.46	2 1/2	5.22	5 1/4

Table 4-7 Suitable Connection Hardware Sizes for Polyester Roundslings, When Used in a Basket Hitch*

Roundslings		Hardware Size			
WSTDA Roundslings Size	Rated Capacity - Basket Hitch (Lbs.)	Minimum Stock Diameter or Thickness (Inches)	Minimum Stock Diameter or Thickness (Inches)* ²	Minimum Effective Contact Width* ³ (Inches)	Minimum Effective Contact Width* ³ (Inches) * ²
1	5,200	.54	9/16	1.37	1 3/8
2	10,600	.83	7/8	1.82	1 7/8
3	16,800	1.02	1 1/16	2.34	2 3/8
4	21,200	1.20	1 1/4	2.52	2 1/2
5	26,400	1.35	1 3/8	2.80	2 7/8
6	33,600	1.59	1 5/8	3.00	3
7	42,400	1.63	1 5/8	3.71	3 3/4
8	50,000	1.77	1 7/8	4.00	4
9	62,000	2.00	2	4.45	4 1/2
10	80,000	2.26	2 3/8	5.06	5
11	106,000	2.69	2 3/4	5.62	5 5/8
12	132,000	2.90	3	6.50	6 1/2
13	180,000	3.50	3 1/2	7.38	7 3/8

*The values in Table 4.7 apply to the use of roundslings in a basket hitch when the two ends of the sling are attached to a single connection point. Use table 4.6 when roundslings are used in a basket hitch when the two ends of the sling are attached to separate connection points.

*² These values are rounded up the closest fractional equivalent.

*³ These values also equal the approximate natural flattening width of the roundslings.

4.7.1.1 Effective Contact Width between the Sling and Connection Hardware

- a. Connection to Flat-Bottom Surfaced Hardware – Such hardware connections include pins, bolts and trunnions. The value of the effective contact width is equal to the opening width or spread of the sling at the connection area (See Figure 4-7). Please note, however, that the effective contact width will never exceed the natural flattening width of the sling as listed in Tables 4-6 and 4-7.
- b. Connection to Round-Bottom (or Curved) Surfaced Hardware – Such hardware connections include links, hooks, or the bow ends of shackles. To determine the value of the effective contact width, multiply the inside opening width of the hardware by a factor of .75 (See Figure 4-8). For connections to the base of hooks, multiply the value of the radius at the hook base by a factor of 1.5 to determine the effective contact

width. Please note, however, that the effective contact width will never exceed the natural flattening width of the sling as listed in Tables 4-6 and 4-7.

STRAIGHT BEARING SURFACE

EFFECTIVE INSIDE WIDTH
EQUALS 100% OF THE
ACTUAL INSIDE WIDTH

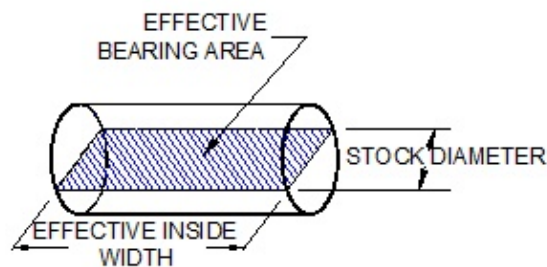
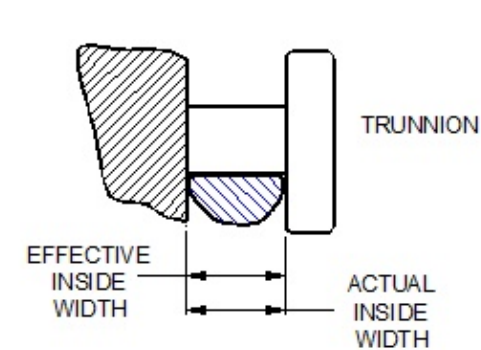


Figure 4-7

CURVED BEARING SURFACE

EFFECTIVE INSIDE WIDTH
EQUALS 75% OF THE
ACTUAL INSIDE WIDTH

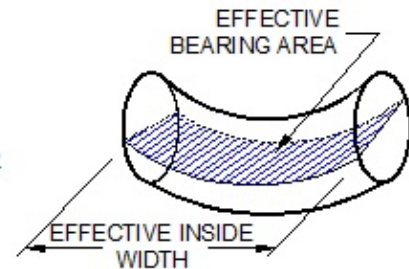
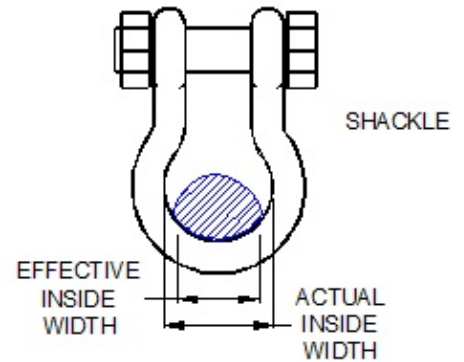


Figure 4-8

Note:

Roundsling strength is affected by the size of the connection hardware. For special applications wherein a retained design factor of 5 is required to be maintained, contact the sling manufacturer, as a capacity reduction of 20% may be appropriate in order to satisfy this criteria.